Sum^k

Statement

Given an array A of N integers $a_1, a_2 \ldots, a_n$. Define the score of a subset S of A as the sum of the elements in S raised to the power of K. That is, for a subset $S = \{s_1, s_2, \ldots, s_m\}$, the score of S is $(s_1 + s_2 + \ldots + s_m)^K$. Find the sum of scores over all possible non-empty subsets of A modulo 998244353.

X modulo M is the remainder when X is divided by M.

Input Format

- The first line consists of two integers N and K.
- Then N integers follow: $a_1, a_2 \ldots, a_n$ $(1 \le a_i \le MAXA)$.

The bounds and points of each test case are given in the following table.

Task	N	K	MAXA	Points
1	10	1	100	5
2	10	2	100	5
3	18	2	10^9	10
4	1000	2	10^9	10
5	100000	2	10^9	10
6	100000	3	10^9	10
7	200	200	10^9	25
8	777	150	10^9	25

Output Format

Output a single integer, the sum of scores over all subsets of A modulo 998244353.

Sample Input 1

```
3 2
1 2 3
```

Sample Output 1

```
100
```

Explanation

There are 7 possible non empty subsets: $\{1\}$, $1^2 = 1$ $\{2\}$, $2^2 = 4$ $\{3\}$, $3^2 = 9$ $\{1, 2\}$, $3^2 = 9$ $\{1, 3\}$, $4^2 = 16$ $\{2, 3\}$, $5^2 = 25$ $\{1, 2, 3\}$, $6^2 = 36$ Totaling up, $1 + 4 + 9 + 9 + 16 + 25 + 36 = 100$.

Sample Input 2

```
2 1
3 3
```

Sample Output 2

```
12
```